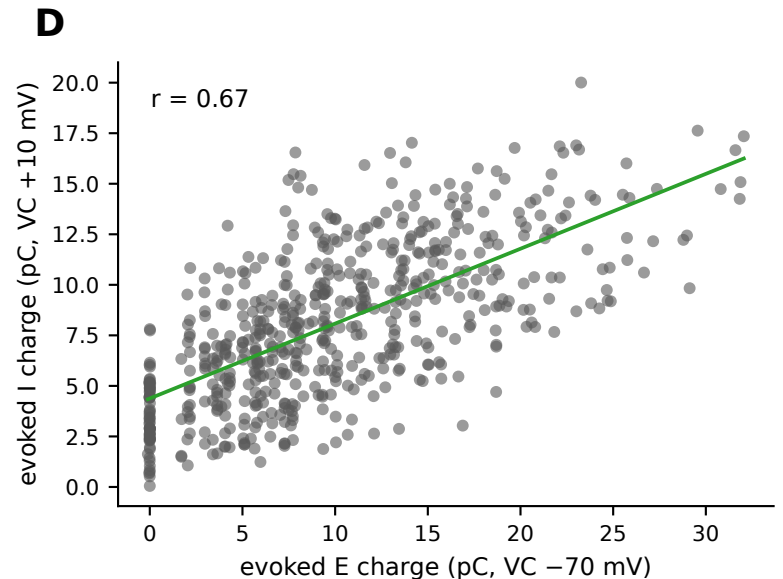
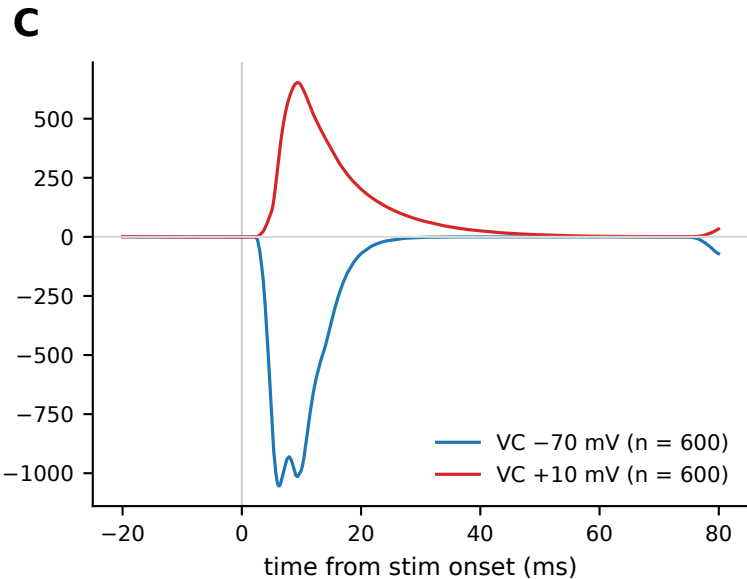
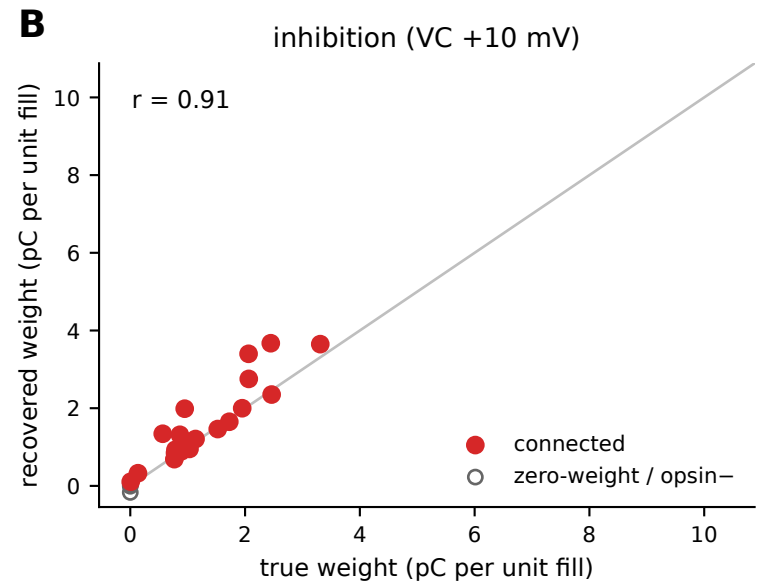
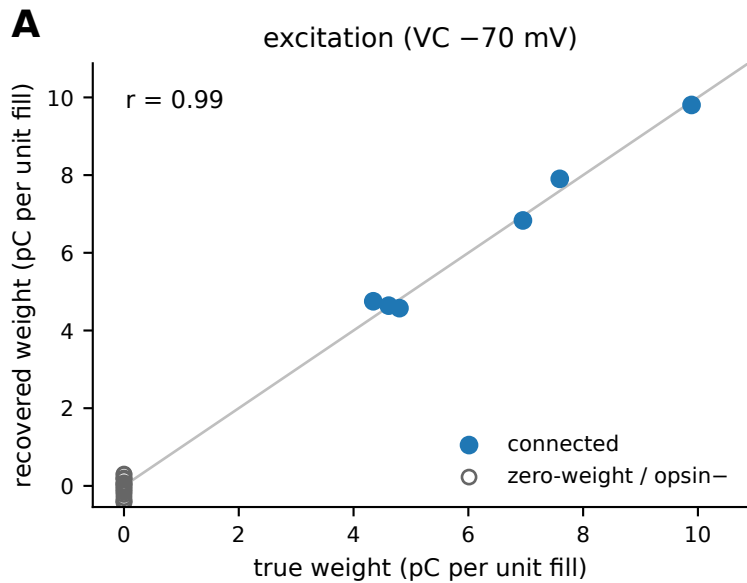




Mock-roundtrip validation: known synaptic weights are recovered from a simulated ensemble E/I mapping session. (A) Recovered vs ground-truth excitatory weight per candidate cell (pC of evoked charge per unit fill fraction), ridge regression on the 600-ensemble fill-fraction design matrix at -70 mV holding; filled symbols are truly connected cells, open symbols zero-weight or opsin- cells, gray line is unity; the never-stimulated patched cell is excluded. **(B)** Same for the inhibitory pathway at $+10$ mV holding. Panels A and B share identical axis limits. **(C)** Grand-average baseline-subtracted evoked current across all ensemble trials, aligned to stimulus onset: net inward at -70 mV (excitation) and net outward at $+10$ mV (inhibition). **(D)** Per-ensemble evoked E charge vs evoked I charge, paired by ensemble across the two holding blocks (sign-corrected so larger means more input; charge integrated 2-60 ms after onset, baseline = pre-onset median), with the linear I~E fit (green). Session mock_session_20260816, seed 42.